

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, IX: EXCEPTIONAL ZEROS, THE GRAPH $\mathcal{L}$ -INVARIANT, AND THE OPERATOR-ALGEBRAIC MAZUR–TATE–TEITELBAUM FORMULA

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ABSTRACT. Section 3 of [III] recorded, without proof, that the cleared voltage pencil $g(T) = T^2 h(T)$ of an abelian $\mathbb{Z}/\ell^k$ tower over a $(q+1)$ -regular graph has a double zero at the trivial character, and that the constant term of the Iwasawa growth law obeys $\nu_0 = \text{ord}_\ell \kappa(Y_0) - \mu + \sum_j \varepsilon_j$ in all four towers of [II, Table 2]. This paper turns both observations into theorems and identifies the number they conceal. The double zero is forced by the symmetry $D(t^{-1}) = D(t)^t$ of the pencil; its second-order coefficient has the closed form

$$\frac{1}{2}\Theta''(0) = -\kappa(Y_0) \|h_\alpha\|^2,$$

where h_α is the harmonic representative of the voltage cochain α — its orthogonal projection onto the cycle space, equivalently onto the kernel of the Hecke–Dirac operator of [II] — and $\kappa(Y_0) = |\text{Jac}(Y_0)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ is the algebraic L -value of [I]. We therefore define the graph $\mathcal{L}$ -invariant $\mathcal{L}(Y_\bullet) := \frac{1}{2}\Theta''(0)/\kappa(Y_0) = -\|h_\alpha\|^2$, a rational number depending only on the class $[\alpha] \in H^1(Y_0; \mathbb{Z})$ and not on ℓ ; for a single-chord voltage, $-\frac{1}{2}\Theta''(0)$ counts the spanning trees avoiding the chord. The bookkeeping identity of [III] becomes a theorem with an exact description of the finitely supported defects ε_j , and the specification’s proposed alignment is replaced by the correct one: in the exceptional-zero regime $\lambda_h = 0$ all defects vanish and $\nu_0 = -\text{ord}_\ell \mathcal{L}(Y_\bullet)$, while in general $\nu_0 + \text{ord}_\ell \mathcal{L}(Y_\bullet) = (\text{ord}_\ell a_2 - \mu) + \sum_j \varepsilon_j$. On the operator-algebraic side we prove a twisted Bass identity for the character-twisted Satake dynamics of [I] and show that at the critical inverse temperature $\beta = 1$ the twisted Bass determinant equals $q^{-n}\Theta(t-1)$: its zero at the trivial character is simple in the temperature direction with derivative proportional to $\kappa(Y_0)$, of order two in the character direction with second derivative proportional to $\mathcal{L}(Y_\bullet)\kappa(Y_0)$, and the partition function of the critical KMS family has a double pole with leading coefficient $n(q - q^{-1})/\mathcal{L}(Y_\bullet)$ — the Mazur–Tate–Teitelbaum and Greenberg–Stevens shapes, with the roles of algebraic L -value and $\mathcal{L}$ -invariant made explicit. The abelian shadow limit of [VIII] carries Θ as the characteristic element of its equivariant K_0 . For the non-normal Iwahori tower there is no character direction: its zero is simple at every level, and the constancy of its residue $n_k \kappa(Y_k) q^{-n_k}$ along the tower is exactly the pseudo-Iwasawa law, with the Euler term $-k$ appearing as the p -adic valuation of the vertex count. No $\mathcal{L}$ -invariant exists for that tower, and we say so.

STATUS OF THE RESULTS

Every numbered statement is proved. The proofs use only results of Papers I–VIII, finite-dimensional linear algebra (a Schur complement replaces analytic perturbation theory), the Weierstrass preparation theorem for $\mathbb{Z}_\ell[[T]]$ [5], and the character product formula for abelian covers already used in [II, §5] and [IV]. The twisted Bass identity is the Artin–Ihara L -function formula of Stark–Terras [4]; we give a six-line proof so that nothing is quoted. Every identity was machine-verified exactly on the four abelian towers of [II, Table 2] — growth laws recomputed directly from

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Paper IX of the series; Papers I–VIII are cited as [I]–[VIII]. The verification script `exceptional_zero.py` in the ancillary bundle reproduces every computational claim; its output is archived as `exceptional_zero.output.txt`.

the reduced Laplacians of the derived graphs up to 128 vertices — and on the Iwahori tower at levels $k \leq 3$ (Table 1). The Mazur–Tate–Teitelbaum [3] and Greenberg–Stevens [2] formulas are cited for the shape of the analogy only; nothing here depends on them. Section 5 records what is open and five corrections to the specification that prompted this paper.

1. THE EXCEPTIONAL ZERO OF THE VOLTAGE PENCIL

Setting as in [II, §5] and [IV]. Y_0 is a finite connected $(q+1)$ -regular graph with n vertices and $m = \frac{(q+1)n}{2}$ edges, $r = m - n + 1$; fix an orientation E^+ of the edges and a voltage $\alpha: E^+ \rightarrow \mathbb{Z}$, extended to oriented edges by $\alpha(\bar{e}) = -\alpha(e)$, so that α is an integer 1-cochain. The voltage adjacency and Laplacian pencils are

$$A(t)_{uv} = \sum_{e: u \rightarrow v} t^{\alpha(e)}, \quad D(t) = (q+1)I_n - A(t) \in M_n(\mathbb{Z}[t, t^{-1}]),$$

and $\Theta(T) := \det D(1+T) \in \mathbb{Z}[T, (1+T)^{-1}] \subset \Lambda := \mathbb{Z}_\ell[[T]]$ is the graph ℓ -adic L -function, the characteristic element of the limit module of [IV, Thm. 2.1]. Its cleared form $g(T) = (1+T)^d \Theta(T) \in \mathbb{Z}[T]$, d minimal, is the polynomial of [II, Rem. 5.4] and [III, §3]; since $(1+T)^d = 1 + dT + \dots$ and $\Theta(0) = \Theta'(0) = 0$ (Theorem 1.2), the coefficients a_i of T^i in Θ and in g agree for $i \leq 2$. We write $\kappa_0 := \kappa(Y_0) = |\text{Jac}(Y_0)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$, the algebraic L -value of [I, Thm. 3.5], and $\Delta_0 = D(1)$ for the Laplacian of Y_0 .

Hodge data. Let $\partial: \mathbb{Z}^{E^+} \rightarrow \mathbb{Z}^V$, $\partial e = t(e) - o(e)$, be the boundary map, so $\Delta_0 = \partial \partial^t$ [II, Lem. 3.2]; the cycle space $\ker \partial = H_1(Y_0; \mathbb{Z})$ and the cut space $\text{im } \partial^t$ are orthogonal complements in $\mathbb{Q}^{E^+}$. With Δ_0^+ the Moore–Penrose inverse (the inverse of Δ_0 on $\mathbf{1}^\perp$, zero on $\mathbf{1}$), the orthogonal projection onto the cycle space is

$$\Pi := I - \partial^t \Delta_0^+ \partial,$$

which is also the projection onto $\ker D_p \cap \ell^2(E^+)$ for the Hecke–Dirac operator D_p of [II, Def. 3.1]. Put $h_\alpha := \Pi \alpha$, the harmonic representative of α , and $\|h_\alpha\|^2 = \alpha^t \Pi \alpha = \|\alpha\|^2 - (\partial \alpha)^t \Delta_0^+ (\partial \alpha) \in \mathbb{Q}_{\geq 0}$.

Lemma 1.1 (Symmetry and switching). *$D(t^{-1}) = D(t)^t$, hence $f(t) := \det D(t)$ satisfies $f(t^{-1}) = f(t)$. For $\varphi \in \mathbb{Z}^V$, the switched voltage $\alpha + \partial^t \varphi$ has pencil $\text{diag}(t^{\varphi_u}) D(t) \text{diag}(t^{-\varphi_u})$, so f depends only on the class $[\alpha] \in H^1(Y_0; \mathbb{Z})$.*

Proof. $A(t^{-1})_{uv} = \sum_{e: u \rightarrow v} t^{-\alpha(e)} = \sum_{e: v \rightarrow u} t^{\alpha(e)} = A(t)_{vu}$. Switching multiplies the entry $t^{\alpha(e)}$, $e: u \rightarrow v$, by $t^{\varphi_v - \varphi_u}$. $\square$

Theorem 1.2 (Exceptional zero of order two). (i) $\Theta(0) = 0$ and $\Theta'(0) = 0$.

(ii) $a_2 = \frac{1}{2} \Theta''(0) = -\kappa(Y_0) \|h_\alpha\|^2$.

(iii) *The zero has order exactly two if and only if $h_\alpha \neq 0$, i.e. if and only if $[\alpha] \neq 0$ in $H^1(Y_0; \mathbb{Q})$.*

(iv) *If α is the indicator of a single edge e , then $\|h_\alpha\|^2 = 1 - R_{\text{eff}}(e)$ and $-a_2 = \kappa(Y_0)(1 - R_{\text{eff}}(e))$ is the number of spanning trees of Y_0 not containing e .*

Proof. Write $s = T$ and expand $D(1+s) = \Delta_0 + sD_1 + s^2D_2 + O(s^3)$ with $D_1 = D'(1) = -A'(1) =: -W$ and $D_2 = \frac{1}{2}D''(1) = -\frac{1}{2}A''(1)$. By Lemma 1.1, $A'(1)^t = -A'(1)$, so W is antisymmetric, and $(W\mathbf{1})_u = \sum_{o(e)=u} \alpha(e) - \sum_{t(e)=u} \alpha(e) = -(\partial \alpha)_u$, i.e. $W\mathbf{1} = -\partial \alpha$. Also $\mathbf{1}^t A''(1) \mathbf{1} = \sum_{e \in E^+} [\alpha(\alpha - 1) + \alpha(\alpha + 1)] = 2\|\alpha\|^2$, so $\mathbf{1}^t D_2 \mathbf{1} = -\|\alpha\|^2$.

Decompose $\mathbb{Q}^n = \mathbb{Q}\mathbf{1} \oplus \mathbf{1}^\perp$ orthogonally, with unit vector $\mathbf{1}/\sqrt{n}$. In this decomposition $\Delta_0 = 0 \oplus \Delta'$ with Δ' invertible and $\det \Delta' = \prod_{\mu \neq 0} \mu = n\kappa_0$ [III, eq. (2.2)]. Writing $E := sD_1 + s^2D_2 + O(s^3)$ in block form $\begin{pmatrix} e_{00} & e_{0*} \\ e_{*0} & E_{**} \end{pmatrix}$, the Schur complement gives

$$f(1+s) = \det(\Delta' + E_{**}) \cdot (e_{00} - e_{0*}(\Delta' + E_{**})^{-1}e_{*0}).$$

Here $e_{00} = \frac{1}{n} \mathbf{1}^t E \mathbf{1} = \frac{s}{n} \mathbf{1}^t D_1 \mathbf{1} + \frac{s^2}{n} \mathbf{1}^t D_2 \mathbf{1} + O(s^3) = -\frac{s^2}{n} \|\alpha\|^2 + O(s^3)$, because $\mathbf{1}^t W \mathbf{1} = 0$ by antisymmetry; and $e_{0*}, e_{*0} = O(s)$ with $e_{*0} = \frac{s}{\sqrt{n}} D_1 \mathbf{1}|_{\mathbf{1}^\perp} + O(s^2)$, $e_{0*} = \frac{s}{\sqrt{n}} (D_1^t \mathbf{1})^t|_{\mathbf{1}^\perp} + O(s^2)$; note $D_1 \mathbf{1} = -W \mathbf{1} = \partial \alpha \in \mathbf{1}^\perp$ and $D_1^t \mathbf{1} = W \mathbf{1} = -\partial \alpha$. Since $(\Delta' + E_{**})^{-1} = \Delta_0^+ + O(s)$ on $\mathbf{1}^\perp$,

$$e_{0*}(\Delta' + E_{**})^{-1}e_{*0} = \frac{s^2}{n}(-\partial \alpha)^t \Delta_0^+(\partial \alpha) + O(s^3).$$

Hence $f(1+s) = (n\kappa_0 + O(s)) \cdot \frac{s^2}{n}(-\|\alpha\|^2 + (\partial \alpha)^t \Delta_0^+(\partial \alpha)) + O(s^3) = -\kappa_0 \|\Pi \alpha\|^2 s^2 + O(s^3)$, which is (i) and (ii). (iii): $\Pi \alpha = 0$ iff α lies in the cut space $\text{im } \partial^t \otimes \mathbb{Q}$ iff $[\alpha] = 0$ in $H^1(Y_0; \mathbb{Q})$. (iv): for $\alpha = e$, $\|\Pi e\|^2 = 1 - e^t \partial^t \Delta_0^+ \partial e = 1 - R_{\text{eff}}(e)$, and $\kappa_0 R_{\text{eff}}(e)$ is the number of spanning trees containing e (Kirchhoff). On K_4 : $R_{\text{eff}} = \frac{1}{2}$, $16 \cdot \frac{1}{2} = 8$ trees avoid the chord, and $a_2 = -8$ in both $(1, 0, 0)$ towers (Table 1). $\square$

Definition 1.3 (Graph $\mathcal{L}$ -invariant). $\mathcal{L}(Y_\bullet) := \frac{\Theta''(0)}{2\kappa(Y_0)} = -\|h_\alpha\|^2 = -\|\Pi_{\ker D_p} \alpha\|^2 \in \mathbb{Q}_{\leq 0}$, nonzero iff $[\alpha] \neq 0$ rationally. It depends only on $[\alpha] \in H^1(Y_0; \mathbb{Z})$ and not on ℓ : the same rational number serves every prime. On K_4 the four towers of [II, Table 2] have $\mathcal{L} = -\frac{1}{2}, -1, -\frac{3}{2}, -\frac{1}{2}$ for $(1, 0, 0)$, $(1, 1, 1)$, $(1, 2, 0)$, $(1, 0, 0)$ respectively.

Theorem 1.4 (The graph Mazur–Tate–Teitelbaum formula). $\frac{1}{2}\Theta''(0) = \mathcal{L}(Y_\bullet) \cdot \kappa(Y_0) = \mathcal{L}(Y_\bullet) \cdot \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$: the leading coefficient of the graph ℓ -adic L -function at its exceptional zero is the $\mathcal{L}$ -invariant times the algebraic L -value.

Proof. Theorem 1.2(ii), Definition 1.3, and [I, Thm. 3.5] for the product formula for $\kappa(Y_0)$. $\square$

Remark 1.5 (The analogy, stated as an analogy). For an elliptic curve with split multiplicative reduction at p , the p -adic L -function has an exceptional zero at $s = 1$ and [3] conjectured, [2] proved, $L'_p(E, 1) = \mathcal{L}(E) L(E, 1)/\Omega_E$ with $\mathcal{L}(E) = \log_p q_E / \text{ord}_p q_E$ the $\mathcal{L}$ -invariant of the Tate period. Theorem 1.4 has that shape: a zero forced by a symmetry, an algebraic part that is a product of Frobenius point counts, and a transcendental-looking factor that is in fact a period-type quantity — here the Hodge energy of the class $[\alpha]$, i.e. its length in the harmonic metric of Y_0 . The zero is of order two rather than one because the graph pencil is self-dual under $t \mapsto t^{-1}$ (Lemma 1.1), which kills the first derivative; the analogue in the classical setting would be a functional equation forcing an even order. We claim the shape and nothing more.

2. THE ℓ -ADIC BOOKKEEPING

Fix a prime ℓ with the tower Y_k connected for all k (equivalently $\alpha \neq 0$ in $H^1(Y_0; \mathbb{F}_\ell)$, true in the four towers). Write $h(T) := g(T)/T^2 \in \mathbb{Z}[T]$, so $h(0) = a_2$, and let $h = \ell^\mu P(T)U(T)$ be its Weierstrass decomposition in Λ : P distinguished of degree λ_h , U a unit [5]. Thus $\mu = \min_i \text{ord}_\ell(\text{coefficients of } h)$ and λ_h is the index of the first coefficient attaining it; [II, Rem. 5.4] and [IV, Cor. 2.3] give the Iwasawa invariants of the tower as μ and $\lambda = \lambda(g) - 1 = \lambda_h + 1$. For $j \geq 1$ put

$$N_j := \text{ord}_\ell \prod_{\zeta \text{ prim. } \ell^j} h(\zeta - 1) = \text{ord}_\ell \text{Res}_T(\Phi_{\ell^j}(1+T), h(T)), \quad \varepsilon_j := N_j - (\mu \varphi(\ell^j) + \lambda_h),$$

the exceptional low-level defects of [III, §3].

Lemma 2.1 (Character product). *For every $k \geq 0$, $\kappa(Y_k) = \kappa(Y_0) \ell^{-k} \prod_{\zeta^{\ell^k}=1, \zeta \neq 1} \det D(\zeta)$ [II, §5], and consequently*

$$\text{ord}_\ell \kappa(Y_k) = \text{ord}_\ell \kappa_0 - k + \sum_{j=1}^k (2 + N_j).$$

Proof. $\det D(\zeta) = \zeta^{-d}g(\zeta - 1)$ and $\prod_{\zeta \neq 1} \zeta^{-d}$ is a root of unity; $g(\zeta - 1) = (\zeta - 1)^2 h(\zeta - 1)$ and $\prod_{\zeta \text{ prim. } \ell^j} \ell^j(\zeta - 1) = \pm \Phi_{\ell^j}(1) = \pm \ell$, of valuation 1; group the nontrivial roots by exact order ℓ^j , $1 \leq j \leq k$. $\square$

Theorem 2.2 (The bookkeeping identity of [III, §3]). *Let $\rho_1, \dots, \rho_{\lambda_h}$ be the roots of P (all of positive valuation) and let $j_0 \geq 1$ be least with $1/\varphi(\ell^{j_0}) < \min_i \text{ord}_\ell \rho_i$ ($j_0 = 1$ if $\lambda_h = 0$). Then:*

- (i) $\varepsilon_j = 0$ for all $j \geq j_0$; the defects are finitely supported;
- (ii) for all $k \geq j_0 - 1$,

$$\text{ord}_\ell \kappa(Y_k) = \mu \ell^k + (\lambda_h + 1)k + \nu_0, \quad \nu_0 = \text{ord}_\ell \kappa(Y_0) - \mu + \sum_{j \geq 1} \varepsilon_j;$$

- (iii) in particular $\lambda = \lambda_h + 1 = \lambda(g) - 1$, recovering [II, Rem. 5.4] and [IV, Cor. 2.3], and the constant term is the identity recorded in [III, §3].

Proof. (i) For ζ primitive of order ℓ^j , $\text{ord}_\ell(\zeta - 1) = 1/\varphi(\ell^j)$; if this is smaller than every $\text{ord}_\ell \rho_i$ then $\text{ord}_\ell(\zeta - 1 - \rho_i) = 1/\varphi(\ell^j)$ for all i , so $\text{ord}_\ell h(\zeta - 1) = \mu + \lambda_h/\varphi(\ell^j)$, and summing over the $\varphi(\ell^j)$ primitive roots gives $N_j = \mu\varphi(\ell^j) + \lambda_h$. (ii) By Lemma 2.1, $\text{ord}_\ell \kappa(Y_k) = \text{ord}_\ell \kappa_0 - k + 2k + \sum_{j \leq k} (\mu\varphi(\ell^j) + \lambda_h + \varepsilon_j) = \text{ord}_\ell \kappa_0 + k + \mu(\ell^k - 1) + \lambda_h k + \sum_{j \leq k} \varepsilon_j$, and for $k \geq j_0 - 1$ the last sum is the full sum. (iii) is read off. Machine confirmation (Table 1): the law holds with the predicted $(\mu, \lambda, \nu_0) = (3, 1, 1), (0, 3, 4), (2, 3, 3), (0, 1, 0)$ — the four rows of [II, Table 2] — at every computed level $k \geq 2$, and in fact at every $k \geq 0$ except in the $(1, 2, 0)$ tower, where $(\varepsilon_1, \varepsilon_2) = (-1, +2)$ and the law starts at $k = 2$, exactly as [III, §3] reported. $\square$

Corollary 2.3 (The exceptional-zero regime). *The following are equivalent: $\lambda_h = 0$; $\text{ord}_\ell a_2 = \mu$; $\text{ord}_\ell \mathcal{L}(Y_\bullet) = \mu - \text{ord}_\ell \kappa(Y_0)$. In this regime $\varepsilon_j = 0$ for every $j \geq 1$, the law of Theorem 2.2(ii) holds for every $k \geq 0$, and*

$$\nu_0 = -\text{ord}_\ell \mathcal{L}(Y_\bullet) :$$

the constant term of the Iwasawa law is minus the ℓ -adic valuation of the $\mathcal{L}$ -invariant. This is the case in both $(1, 0, 0)$ towers: $\nu_0 = 1 = -\text{ord}_2(-\frac{1}{2})$ and $\nu_0 = 0 = -\text{ord}_3(-\frac{1}{2})$.

Proof. $\mu \leq \text{ord}_\ell a_2$ always, with equality iff the constant coefficient attains the minimum iff $\lambda_h = 0$; the third condition is the second rewritten. If $\lambda_h = 0$ then $P = 1$, $\text{ord}_\ell h(\zeta - 1) = \mu$ for every ζ , so $N_j = \mu\varphi(\ell^j)$ and $\varepsilon_j = 0$, $j_0 = 1$; then $\nu_0 = \text{ord}_\ell \kappa_0 - \mu = \text{ord}_\ell \kappa_0 - \text{ord}_\ell a_2 = -\text{ord}_\ell \mathcal{L}$. $\square$

Theorem 2.4 (The $\mathcal{L}$ -defect identity). *In general,*

$$\nu_0 + \text{ord}_\ell \mathcal{L}(Y_\bullet) = (\text{ord}_\ell a_2 - \mu) + \sum_{j \geq 1} \varepsilon_j =: \delta(Y_\bullet, \ell),$$

with $\text{ord}_\ell a_2 - \mu \geq 0$ and equality iff $\lambda_h = 0$. On the four towers $\delta = 0, 4, 2, 0$.

Proof. Substitute $\text{ord}_\ell \mathcal{L} = \text{ord}_\ell a_2 - \text{ord}_\ell \kappa_0$ into Theorem 2.2(ii). $\square$

Remark 2.5 (A correction to the specification). The specification asked to prove that $\text{ord}_\ell \mathcal{L}(Y_\bullet)$ “aligns exactly” with $\nu_0 - \text{ord}_\ell \kappa(Y_0) + \mu$. By Theorem 2.2 the latter quantity is $\sum_j \varepsilon_j$, which is $0, 0, 1, 0$ on the four towers, whereas $\text{ord}_\ell \mathcal{L} = -1, 0, -1, 0$: there is no such identity. The statements that hold are Corollary 2.3 and Theorem 2.4. The specification’s other claim, that $\text{ord}_\ell a_2 = \mu$ “controls the main term in the $\lambda_h = 0$ regime”, is the definition of that regime; its content is the vanishing of all defects and the formula $\nu_0 = -\text{ord}_\ell \mathcal{L}$.

Proposition 2.6 (The exceptional zero inside the Iwasawa module). *Let $X_\infty \cong \Lambda^n/D(1+T)\Lambda^n$ be the limit module of [IV, Thm. 2.1], with $\text{char}_\Lambda(X_\infty) = (\Theta)$. Then $X_\infty/TX_\infty \cong (\mathbb{Z} \oplus \text{Jac}(Y_0)) \otimes \mathbb{Z}_\ell$ has $\mathbb{Z}_\ell$ -rank one, while $\text{ord}_T \Theta = 2$; at the augmentation prime the localization is*

$$X_\infty \otimes_\Lambda \Lambda_{(T)} \cong \Lambda_{(T)}/T^2\Lambda_{(T)},$$

cyclic of length two. The exceptional zero is the statement that the Eisenstein eigenspace of the Iwasawa module is not semisimple.

Proof. The first isomorphism is the exact control theorem [IV, Cor. 2.2] at $k = 0$. $\Lambda_{(T)}$ is a discrete valuation ring with uniformizer T and residue field $\mathbb{Q}_\ell$, so $D(1+T)$ has elementary divisors over it; reducing modulo T gives $D(1) = \Delta_0$ of rank $n-1$ over $\mathbb{Q}_\ell$, so exactly one elementary divisor is divisible by T , and since the T -adic order of the determinant is 2 (Theorem 1.2) that divisor is T^2 times a unit. $\square$

3. THE OPERATOR-ALGEBRAIC FORMULATION

Setting from [I]. $\mathcal{G}_\Gamma$ is the Hecke groupoid, $C^*(\mathcal{G}_\Gamma) = \mathcal{O}_{A^{\text{nb}}}$ the Cuntz–Krieger algebra of the non-backtracking matrix $A^{\text{nb}} = T^t S - J$ [I, Def. 1.7, Prop. 1.9(ii)], where $S, T \in M_{n \times 2m}(\mathbb{Z})$ are the start and terminal incidence matrices of oriented edges and J is edge reversal. The Satake dynamics $\sigma^{(\pi)}$ [I, Def. 2.1] has partition function $Z_p(\beta; \pi) = L_p(\beta, \pi_p)$ [I, Thm. 2.2]; for the Eisenstein parameter $(\alpha_1, \alpha_2) = (1, q)$ the KMS_β simplex is nonempty exactly at the critical inverse temperature $\beta = 1$, where it is a single Perron state [I, Thm. 2.4(ii),(iv)]; off the Perron locus the Satake data survive only as KMS *functionals* [I, Thm. 2.5, Rem. 2.6]. Throughout $u = q^{-\beta}$, so $\beta = 1$ is $u = q^{-1}$.

Definition 3.1 (Character twist). The voltage α is a $\mathbb{Z}$ -valued continuous 1-cocycle c_α on $\mathcal{G}_\Gamma$ — the sum of α along the edges of a Hecke move, well defined on $(\omega, i - j, \omega')$ because the tails agree — commuting with the degree cocycle c . It induces a $\mathbb{T}$ -action γ^α on $C^*(\mathcal{G}_\Gamma)$ commuting with $\sigma^{(\pi)}$, and for $t \in \mathbb{C}^\times$ the *twisted transfer operator* and *twisted Hecke operators*

$$B_t := B\Lambda_t, \quad \Lambda_t := \text{diag}(t^{\alpha(e)})_e, \quad T_k(t)_{uv} := \sum_{\text{nb paths } u \rightarrow v \text{ of length } k} t^{\alpha(\text{path})},$$

with $B = A^{\text{nb}}$ on $\mathbb{C}^{2m}$; for $|t| = 1$ these are the character-twisted Ruelle–Hecke transfer operator of [I, Def. 2.1] at the Eisenstein potential and the twisted Hecke correspondences. Note $T_1(t) = A(t)$ and $\Lambda_t J \Lambda_t = J$, since $\alpha(\bar{e}) = -\alpha(e)$.

Lemma 3.2 (Twisted Hecke recursion). $A(t)T_k(t) = T_{k+1}(t) + qT_{k-1}(t)$ for $k \geq 2$, $A(t)^2 = T_2(t) + (q+1)I$, hence

$$\sum_{k \geq 0} T_k(t) u^k = (1 - u^2)(I - uA(t) + qu^2 I)^{-1}$$

as an identity of matrix-valued rational functions of (u, t) .

Proof. $A(t)T_k(t)$ sums over an edge $e: u \rightarrow v$ followed by a non-backtracking path $w \rightarrow v$ of length k , with weight $t^{\alpha(e)} t^{\alpha(\text{path})}$. The terms in which the path begins with $\bar{e}$ have weight $t^{\alpha(e)+\alpha(\bar{e})} t^{\alpha(\text{rest})} = t^{\alpha(\text{rest})}$, the rest being a non-backtracking path $u \rightarrow v$ of length $k-1$ whose first edge is not e ; for each such rest there are q admissible e . This is the proof of [I, Lem. 1.5] with the multiplicative weight carried along; the generating function follows by the same algebra. $\square$

Theorem 3.3 (Twisted Bass identity). With $P(u, t) := \det(I_n - uA(t) + qu^2 I_n)$,

$$\det(I_{2m} - uB\Lambda_t) = (1 - u^2)^{r-1} P(u, t).$$

Proof. Recall $S\Lambda_t T^t = A(t)$ (entry (u, v) sums $t^{\alpha(e)}$ over $e: u \rightarrow v$), $SJ = T$, $TT^t = (q+1)I$, and $\Lambda_t J \Lambda_t = J$. The operator $J\Lambda_t$ is an involution ($(J\Lambda_t)^2 = J\Lambda_t J \Lambda_t = JJ = I$) with trace 0 (no edge is its own reverse), so its eigenvalues are ± 1 with multiplicity m each and $\det(I + uJ\Lambda_t) = (1 - u^2)^m$, with $(I + uJ\Lambda_t)^{-1} = (1 - u^2)^{-1}(I - uJ\Lambda_t)$. Then $I - uB\Lambda_t = (I + uJ\Lambda_t) - uT^t S\Lambda_t$, and by

$$\det(I - XY) = \det(I - YX),$$

$$\begin{aligned} \det(I - uB\Lambda_t) &= (1 - u^2)^m \det\left(I_n - \frac{u}{1-u^2} S\Lambda_t(I - uJ\Lambda_t)T^t\right) \\ &= (1 - u^2)^{m-n} \det\left((1 - u^2)I_n - u(A(t) - uSJT^t)\right) \\ &= (1 - u^2)^{m-n} \det(I - uA(t) + qu^2I), \end{aligned}$$

using $S\Lambda_t J\Lambda_t T^t = SJT^t = TT^t = (q+1)I$ and $m - n = r - 1$. For $t = 1$ this is Bass's identity [I, Prop. 3.4]; for $|t| = 1$ it is the Artin–Ihara L -function formula of [4]. Verified as a polynomial identity in (u, t) on all four voltage assignments (Table 1). $\square$

Theorem 3.4 (Exceptional zero at the critical temperature). *Let $P(u, t)$ be as above and $Z(\beta; t) := \text{tr} \sum_k T_k(t) q^{-k\beta}$ the twisted partition function of the Eisenstein Satake dynamics.*

- (i) *Functional equation: $P(\frac{1}{qu}, t) = (qu^2)^{-n} P(u, t)$.*
- (ii) *At the critical inverse temperature $\beta = 1$, $P(q^{-1}, t) = q^{-n} \Theta(t - 1)$: the twisted Bass determinant of the Satake dynamics at $\beta = 1$ is the graph ℓ -adic L -function.*
- (iii) *At $(u, t) = (q^{-1}, 1)$ the zero is simple in the temperature direction and of order two in the character direction:*

$$\begin{aligned} \partial_u P(q^{-1}, 1) &= -n(q - 1) q^{1-n} \kappa(Y_0), \\ \partial_t P(q^{-1}, 1) &= 0, \quad \frac{1}{2} \partial_t^2 P(q^{-1}, 1) = q^{-n} \mathcal{L}(Y_\bullet) \kappa(Y_0). \end{aligned}$$

The ratio $\frac{1}{2} \partial_t^2 P / \partial_u P = -\mathcal{L}(Y_\bullet) / (nq(q - 1)) = \|h_\alpha\|^2 / (nq(q - 1))$ is independent of $\kappa(Y_0)$.

- (iv) *The unique KMS_1 state of [I, Thm. 2.4] is the $t = 1$ member of the family; for $t \neq 1$ on the unit circle with $t^{\alpha(C)} \neq 1$ for some cycle C there is no KMS_1 state, only the functional of [I, Rem. 2.6], and the critical partition function has a double pole at the trivial character with*

$$Z(1; t) = q(1 - q^{-2}) \text{tr} D(t)^{-1} = \frac{n(q - q^{-1})}{\mathcal{L}(Y_\bullet)} \frac{1}{(t - 1)^2} + O((t - 1)^{-1}).$$

Proof. (i) $I - \frac{1}{qu}A + \frac{q}{q^2u^2}I = (qu^2)^{-1}(qu^2I - uA + I)$. (ii) At $u = q^{-1}$, $I - q^{-1}A(t) + q^{-1}I = q^{-1}D(t)$. (iii) $\partial_t P(q^{-1}, t) = q^{-n} f'(t)$ and $\frac{1}{2} \partial_t^2 P(q^{-1}, 1) = q^{-n} a_2$, so Theorem 1.2 gives the t -derivatives. For the u -derivative, $\partial_u \det M(u) = \text{tr}(\text{adj} M(u) M'(u))$ with $M(q^{-1}) = q^{-1} \Delta_0$, $\text{adj}(q^{-1} \Delta_0) = q^{1-n} \text{adj} \Delta_0 = q^{1-n} \kappa_0 \mathbf{1} \mathbf{1}^t$ (all cofactors of a Laplacian equal κ_0), and $M'(q^{-1}) = -T_p + 2I$, so $\partial_u P = q^{1-n} \kappa_0 \mathbf{1}^t (2I - T_p) \mathbf{1} = q^{1-n} \kappa_0 n(2 - (q+1))$. (iv) Existence of a KMS_1 state requires $\rho(q^{-1}B\Lambda_t) = 1$ [I, Thm. 2.4(i)]; for $|t| = 1$ the twisted Laplacian $D(t)$ is Hermitian positive semidefinite with $D(t)v = 0$ forcing $|v|$ constant and $t^{\alpha(C)} = 1$ on every cycle, so under the hypothesis $D(t)$ is positive definite, $P(q^{-1}, t) \neq 0$, and the pressure equation fails (Perron obstruction, as in [I, Thm. 2.5]). By Lemma 3.2, $Z(1; t) = (1 - q^{-2}) \text{tr}(I - q^{-1}A(t) + q^{-1}I)^{-1} = q(1 - q^{-2}) \text{tr} D(t)^{-1}$. The Perron eigenvalue branch of $D(t)$ through 0 at $t = 1$ is $\lambda(1 + s) = -\frac{\|h_\alpha\|^2}{n} s^2 + O(s^3)$ (this is the Schur-complement computation in the proof of Theorem 1.2: $f(1 + s)/\det \Delta'$), and the other $n - 1$ eigenvalues are bounded away from zero, so $\text{tr} D(t)^{-1} = -\frac{n}{\|h_\alpha\|^2} (t - 1)^{-2} + O((t - 1)^{-1}) = \frac{n}{\mathcal{L}(Y_\bullet)} (t - 1)^{-2} + O((t - 1)^{-1})$. On K_4 the leading coefficients are $-12, -6, -4, -12$ (Table 1). $\square$

Remark 3.5 (Which derivative). The specification asked for “the second derivative of the KMS functional at $\beta = 1$ ”. Theorem 3.4 locates the derivatives precisely: at the critical point the *temperature* derivative is simple and carries the algebraic L -value $\kappa(Y_0)$; the *character* derivative is of second order and carries $\mathcal{L}(Y_\bullet) \kappa(Y_0)$; their ratio is the $\mathcal{L}$ -invariant up to the explicit normalization $nq(q - 1)$, with $\kappa(Y_0)$ cancelling — which is the Greenberg–Stevens shape [2], a ratio of a derivative in the “weight” direction to one in the cyclotomic direction. The specification’s request for a cyclic-cohomology formulation is not met here: the natural candidate pairs the derivation generating γ^α

with the Fredholm module of [II, Thm. 3.3], and we have not established that pairing. We record it as open (Remark 5.1) rather than dress the second derivative in cohomological language it has not earned.

Theorem 3.6 (The abelian shadow limit carries Θ). *Let (Y_k) be a connected abelian $\mathbb{Z}/\ell^k$ voltage tower, $\mathcal{O}_k := \mathcal{O}_{B(Y_k)}$ the shadow algebras of [VI] ($c = q$ at every level, all levels being $(q+1)$ -regular), and $\pi: Y_{k+1} \rightarrow Y_k$ the covering maps. Then π extends to an in-covering $\pi^\sharp$ of shadow graphs, [VIII, Lem. 2.1] yields unital injective $*$ -homomorphisms $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$, and $\mathcal{O}_\infty^{\text{ab}} := \varinjlim (\mathcal{O}_k, \varphi_k)$ is a unital Kirchberg algebra with*

$$K_1(\mathcal{O}_\infty^{\text{ab}}) \cong \mathbb{Z}, \quad \text{Tor } K_0(\mathcal{O}_\infty^{\text{ab}}) \cong \text{Hom}_{\text{cont}}\left(\varprojlim_k (\text{Jac}(Y_k), \pi_*), \mathbb{Q}/\mathbb{Z}\right), \quad K_0/\text{Tor} \cong \mathbb{Z}[1/\ell].$$

The deck transformations induce a continuous action of $\mathbb{Z}_\ell = \varprojlim \mathbb{Z}/\ell^k$ on $\mathcal{O}_\infty^{\text{ab}}$ commuting with the gauge action, and the ℓ -primary torsion of K_0 is, as a discrete Λ -module, the Pontryagin dual of the compact Λ -module $\text{Tor } X_\infty$ of [IV, Thm. 2.1], whose characteristic ideal is (Θ) . The ideal (Θ) is stable under the involution $1 + T \mapsto (1 + T)^{-1}$ (Lemma 1.1), so the same Θ — with its exceptional zero and its $\mathcal{L}$ -invariant — is the characteristic element on both sides of the duality.

Proof. A covering map is bijective on in-arcs and on out-arcs at every vertex, so $\pi^\sharp$ ($v \mapsto \pi(v)$, $v' \mapsto \pi(v)'$, shadow arcs label-wise) is an in-covering, and [VIII, Lem. 2.1, Thm. 2.3] apply verbatim with τ replaced by π : $K_0(\varphi_k)$ is the pullback π^* on $\text{coker } \Delta_k$ (the Laplacians are symmetric, so the transpose convention of [VIII, Conv. 1.1] is invisible), which is the Pontryagin adjoint of the norm π_* of [IV, Lem. 1.2–1.3] by [VIII, Lem. 1.5]; $\pi^* \mathbf{1} = \mathbf{1}$ gives unitality, $\deg \pi^* = \ell \deg$ gives the free part, and $K_1(\varphi_k) = \text{id}$ as in [VIII, Thm. 2.3(iii)]. The limit is Kirchberg by [VIII, Thm. 3.2] and its K -theory follows as in [VIII, Thm. 3.3]. Each deck generator σ_k is an automorphism of $Y_k^\sharp$ covering σ_{k-1} , hence induces an automorphism of $\mathcal{O}_k$ with $\sigma_{k+1} \circ \varphi_k = \varphi_k \circ \sigma_k$ (both send p_v to $\sum_{\pi(w)=\sigma_k v} p_w$), so $\mathbb{Z}_\ell$ acts on the limit, continuously because on each $\mathcal{O}_k$ the action factors through $\mathbb{Z}/\ell^k$. On $K_0(\mathcal{O}_k) = \text{coker } \Delta_k \otimes \mathbb{Z}_\ell$ the deck action is multiplication by $\sigma \in \mathbb{Z}_\ell[G_k]$ [IV, Lem. 1.1], so the ℓ -part of $\text{Tor } K_0(\mathcal{O}_\infty^{\text{ab}}) = \varinjlim \text{Jac}(Y_k)^\vee$ is the Pontryagin dual of $\varprojlim \text{Jac}(Y_k) \otimes \mathbb{Z}_\ell = \text{Tor } X_\infty$ with the contragredient Λ -action, and $\text{char}_\Lambda(X_\infty) = (\Theta)$ is [IV, Thm. 2.1]. The involution stability is $f(t^{-1}) = f(t)$. $\square$

Remark 3.7 (Two algebras, one zero). The shadow algebras are Hecke-functorial and not Hecke-dynamical [VI, Rem. 4.3]: their gauge dynamics reach the Laplacian at $u = 1$, i.e. $\beta = 0$, where pure infiniteness forbids traces and hence KMS_0 states. So the exceptional zero has two operator-algebraic homes, and they do not overlap: the *dynamical* side — partition functions, the critical temperature $\beta = 1$, the double pole of Theorem 3.4(iv) — lives on $C^*(\mathcal{G}_\Gamma)$ of [I]; the *K-theoretic* side — Θ as characteristic element, Proposition 2.6 — lives on the equivariant limit $\mathcal{O}_\infty^{\text{ab}}$ of Theorem 3.6. The specification's request to read the derivative off “the gauge action of $\mathcal{O}_\infty$ ” conflates the two; by [II, Thm. 1.6] no single algebra can carry both.

4. THE IWAHORI TOWER: A SIMPLE ZERO AND NO $\mathcal{L}$ -INVARIANT

Theorem 4.1 (Residue constancy along the Iwahori tower). *For the Iwahori path tower $(Y_k)_{k \geq 1}$ of [II, §5] over a $(q+1)$ -regular base with $q = p$, let $\zeta_k(u)^{-1} := \det(I - uA_k)$. Then:*

- (i) $\zeta_k(u)^{-1}$ is independent of $k \geq 1$ and equals $\det(I - uA^{\text{nb}}) = (1 - u^2)^{r-1} P(u, 1)$;
- (ii) $u = q^{-1}$ is a simple zero of $\zeta_k(u)^{-1}$ at every level, with

$$\frac{d}{du} \det(I - uA_k) \Big|_{u=q^{-1}} = -q^{2-n_k} n_k \kappa(Y_k);$$

- (iii) consequently $n_k \kappa(Y_k) q^{-n_k}$ is independent of $k \geq 1$, which is the recursion $\kappa(Y_{k+1}) = q^{n_k(q-1)-1} \kappa(Y_k)$ of [III, Thm. 2.1], and comparing with level 0,

$$\kappa(Y_1) = \frac{(1 - q^{-2})^{r-1} (q-1) n q^{n_1-1-n}}{n_1} \kappa(Y_0);$$

- (iv) the pseudo-Iwasawa law [III, Thm. 2.2] reads $\text{ord}_p \kappa(Y_k) = n_k - \text{ord}_p n_k + \text{const} = \mu p^k - k + \nu_0$: the Euler term $-k$ is the p -adic valuation of the vertex count $n_k = n(q+1)q^{k-1}$, i.e. of the Eisenstein normalization in the residue.

Proof. (i) $A_1 = A^{\text{nb}}$, and $\det(zI - A_{k+1}) = z^{n_{k+1}-n_k} \det(zI - A_k)$ [III, eq. (2.1)] gives $\det(I - uA_{k+1}) = \det(I - uA_k)$; the last equality is Bass [I, Prop. 3.4]. (ii) With $M(u) = I - uA_k$, $M(q^{-1}) = q^{-1}\Delta_k$ has the simple eigenvalue 0 (Perron), so the zero is simple; $\text{adj}(q^{-1}\Delta_k) = q^{1-n_k} \kappa(Y_k) \mathbf{1}\mathbf{1}^t$ since all cofactors of the Eulerian Laplacian equal $\kappa(Y_k)$ [III, proof of Thm. 2.1, Step 2], and $\text{tr}(\mathbf{1}\mathbf{1}^t(-A_k)) = -\mathbf{1}^t A_k \mathbf{1} = -qn_k$. (iii) Equate the derivatives at consecutive levels, using $n_{k+1} = qn_k$; for the level-0 formula differentiate $(1 - u^2)^{r-1} P(u, 1)$ at q^{-1} using Theorem 3.4(iii). (iv) $\text{ord}_p(n_k \kappa_k q^{-n_k})$ constant gives $\text{ord}_p \kappa_k = n_k - \text{ord}_p n_k + \text{const}$, and $\text{ord}_p n_k = \text{ord}_p(n(q+1)) + k - 1$. On K_4 : $n_k \kappa_k q^{-n_k} = \frac{9}{8}$ at $k = 1, 2, 3$, the derivative is $-\frac{9}{2}$ at every level, and $\kappa(Y_1) = \frac{(3/4)^2 \cdot 4 \cdot 2^7}{12} \cdot 16 = 384$ (Table 1). $\square$

Remark 4.2 (No $\mathcal{L}$ -invariant for the non-normal tower). The specification asked for $\Theta''(0)$, the $\mathcal{L}$ -invariant and the defects ε_j “on the Iwahori path tower as well”. There is no Θ there: the tower carries no deck action, hence no character variable, hence no pencil, and the Hecke variable u is the only direction available. In that direction the zero at the critical temperature is simple at every level (Theorem 4.1(ii)), with residue $\kappa(Y_k)$ rather than a second-order coefficient. What replaces the $\mathcal{L}$ -invariant is the tower constant $n_k \kappa(Y_k) q^{-n_k}$, and what replaces the bookkeeping identity is the observation that its constancy is the growth law. This gives the Euler term $-k$ its third reading, after the Eisenstein rank removed once per level [III, Thm. 2.4] and the Eisenstein index through the connecting map of the snake [VII, Cor. 2.2]: it is the p -adic valuation of the vertex count, the normalization $\prod_{\mu \neq 0} \mu = n_k \kappa_k$ of [III, eq. (2.2)] read p -adically. The three readings are one fact seen from three sides, and none of them is an $\mathcal{L}$ -invariant.

5. SCOPE, RESIDUE, CORRECTIONS

Remark 5.1 (Solved, open, impossible). *Solved*: the direction of [III, §3], made precise. The double zero is a theorem with the closed form $\frac{1}{2} \Theta''(0) = -\kappa(Y_0) \|h_\alpha\|^2$; the $\mathcal{L}$ -invariant is defined, computed, and shown to be a Hodge energy and ℓ -independent; the bookkeeping identity holds with an exact description of the defects, and the constant term is $-\text{ord}_\ell \mathcal{L}$ in the exceptional-zero regime; the twisted Bass identity and the critical-temperature theorem give the Mazur–Tate–Teitelbaum and Greenberg–Stevens shapes on $C^*(\mathcal{G}_T)$; the abelian shadow limit carries Θ as equivariant characteristic element. *Open*: (a) a cyclic-cohomology formulation of Theorem 3.4(iii) (Remark 3.5); (b) a converse to Theorem 2.4 — whether $\delta(Y_\bullet, \ell) = 0$ forces $\lambda_h = 0$; (c) a structural formula for the defects ε_j at $j < j_0$ in terms of the Newton polygon of P at the levels ℓ^j , beyond the resultant definition; (d) everything about the non-normal tower that [V, Prob. 5.1] and [VII, Rem. 3.1] left open, which this paper does not touch. *Impossible or false as stated*: an $\mathcal{L}$ -invariant for the Iwahori tower in the sense of Definition 1.3 (Remark 4.2); the alignment formula of the specification (Remark 2.5); a single algebra carrying both the dynamical and the K -theoretic side (Remark 3.7, by [II, Thm. 1.6]).

Remark 5.2 (Five corrections to the task specification). (1) $\Theta(T) = \det D(1 + T)$ lies in $\mathbb{Z}[T, (1 + T)^{-1}] \subset \Lambda$, not in $\mathbb{Z}_\ell[T]$; the cleared g is the polynomial, and a_2 is the same for both. (2) “ $\text{ord}_\ell a_2 = \mu$ in the $\lambda_h = 0$ regime” is the definition of the regime; the theorem is Corollary 2.3. (3) The alignment “ $\text{ord}_\ell \mathcal{L} \leftrightarrow \nu_0 - \text{ord}_\ell \kappa_0 + \mu$ ” is false; Theorem 2.4 is the identity that holds.

(4) The KMS second derivative is in the character direction, at $\beta = 1$, on $C^*(\mathcal{G}_\Gamma)$ and not on the shadow limit; the temperature derivative carries $\kappa(Y_0)$ (Theorem 3.4, Remarks 3.5–3.7); no cyclic-cohomology statement is claimed. (5) The Iwahori tower has no Θ , no Θ'' , and no ε_j ; Theorem 4.1 is what it has. Separately, [III, Rem. 1.11] and [VIII] cite the tail-groupoid K -theory as [I, Prop. 1.7(iii)]; in the compiled numbering of [I] it is Proposition 1.9(iii), and [VIII] has been corrected accordingly.

6. THE VERIFICATION BATTERY

Table 1 lists the exact checks of `exceptional_zero.py`; the letters match the docstring of the script and the references in the text. The four abelian towers are those of [II, Table 2]; the growth laws were recomputed directly from the reduced Laplacians of the derived graphs ($k \leq 5$ for $\ell = 2$, up to 128 vertices; $k \leq 3$ for $\ell = 3$) and independently from the character product of Lemma 2.1, and the two agree at every level.

check	towers / levels	result
(Z) $\Theta(0) = \Theta'(0) = 0$; $a_2 = -8, -16, -24, -8$; $g = -8T^2, -T^4 - 16T^3 - 16T^2, -4T^4 - 24T^3 - 24T^2$	4 towers	exact
(H) $a_2 = -\kappa_0 \ h_\alpha\ ^2$; $\ h_\alpha\ ^2 = \frac{1}{2}, 1, \frac{3}{2}, \frac{1}{2}$; $\mathcal{L} = -\frac{1}{2}, -1, -\frac{3}{2}, -\frac{1}{2}$	4 towers	exact
(H') single chord: $R_{\text{eff}} = \frac{1}{2}$, 8 spanning trees avoid the chord $= -a_2$	(1, 0, 0)	exact
(K) double pole of $Z(1; t)$: leading coefficient $= n(q - q^{-1})/\mathcal{L} = -12, -6, -4, -12$	4 towers	exact
(M) Newton data $(\mu, \lambda_h) = (3, 0), (0, 2), (2, 2), (0, 0)$; regime iff $\text{ord}_\ell a_2 = \mu$	4 towers	exact
(E) ε_j , $j \leq 5$: all zero except $(\varepsilon_1, \varepsilon_2) = (-1, 2)$ for (1, 2, 0)	4 towers	exact
(G) $\text{ord}_\ell \kappa(Y_k)$ direct = character formula = law; (μ, λ, ν_0) as [II, Table 2]	$k \leq 5 / k \leq 3$	exact
(L) $\nu_0 + \text{ord}_\ell \mathcal{L} = (\text{ord}_\ell a_2 - \mu) + \sum \varepsilon_j = 0, 4, 2, 0$; regime $\Rightarrow \nu_0 = -\text{ord}_\ell \mathcal{L}$	4 towers	exact
(B) $\det(I - uB\Lambda_t) = (1 - u^2)^{r-1}P(u, t)$ as a polynomial identity in (u, t)	4 towers	exact
(F) functional equation; $P(q^{-1}, t) = q^{-n}\Theta(t - 1)$; $\partial_u P = -8$; $\frac{1}{2}\partial_t^2 P = q^{-n}a_2$	4 towers	exact
(S) $\text{rk } D(1) = 3 = n - 1$, $\text{ord}_T \Theta = 2$	4 towers	exact
(I) Iwahori: level-independent $\det(I - uA_k)$; $n_k \kappa_k q^{-n_k} = \frac{9}{8}$; slope $-\frac{9}{2}$ at $u = \frac{1}{2}$; $\kappa(Y_1) = 384$	$k \leq 3$	exact

TABLE 1. Exact verification (`exceptional_zero.py`) on K_4 ($q = 2$, $n = 4$, $r = 3$, $\kappa_0 = 16$). Tower order: (1, 0, 0), $\ell = 2$; (1, 1, 1), $\ell = 2$; (1, 2, 0), $\ell = 2$; (1, 0, 0), $\ell = 3$. For the fixed towers each line is itself a proof; the general statements are proved in the text.

ACKNOWLEDGEMENT OF PROVENANCE AND CAVEATS

Unrefereed preprint. Every numbered statement is proved from Papers I–VIII, finite linear algebra, Weierstrass preparation [5] and the character product formula; the twisted Bass identity is proved rather than quoted from [4]. The Mazur–Tate–Teitelbaum and Greenberg–Stevens references [3, 2] fix the shape of an analogy (Remarks 1.5, 3.5) and are load-bearing for nothing. Remark 5.2 records the corrections to the specification that prompted the paper and one citation correction propagated from [III] to [VIII]. Bibliographic data of [3, 2, 4, 1] were checked against publisher or indexing pages (titles, venues, years, volumes, pages); [5] is the entry verified in [IV].

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